Transition from IPv4 to IPv6

Computer Network Technologies and Services

Tecnologie e Servizi di Rete

Internet Architecture and Technologies

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The road to IPv6

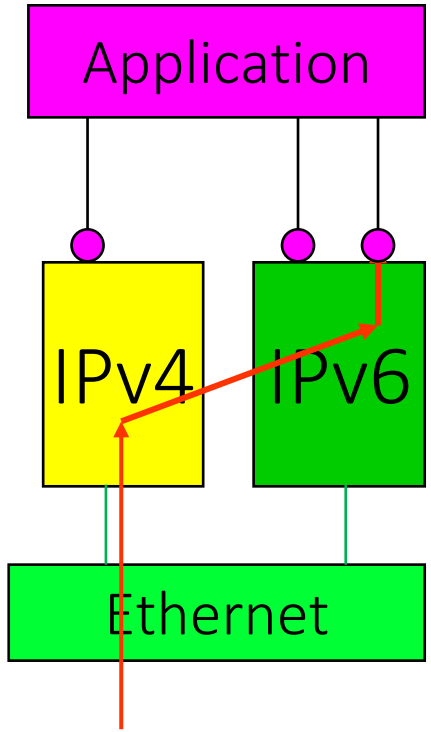


IPv4 to IPv6 transition

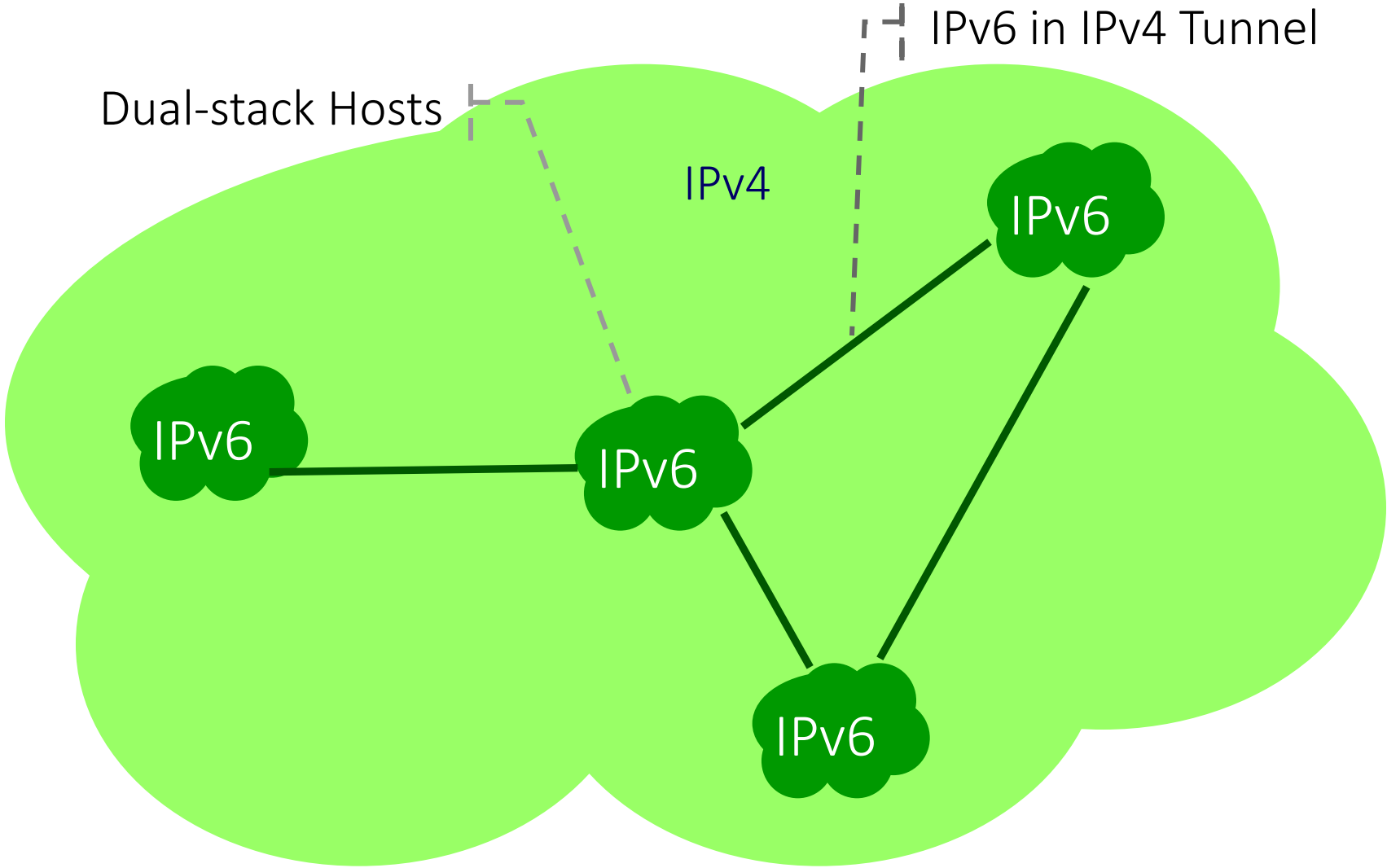
- Ideally:
  - Incremental
  - Seamless
  - Smooth

How can it be achieved?

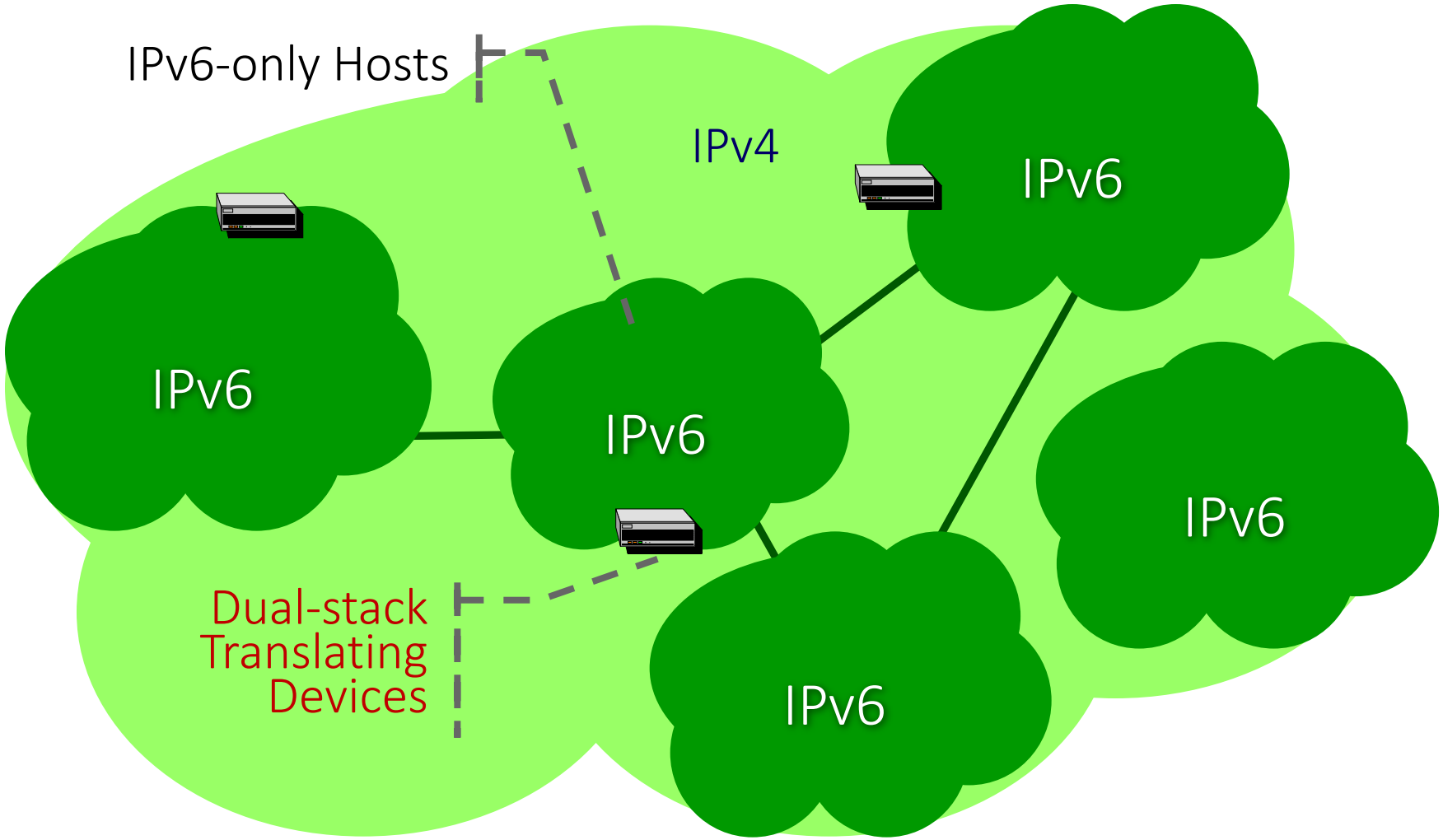
- Dual-stack approach
  - IPv6 as a new layer-3 protocol
  - Generate/receive v6 or v4 packets as needed
- Address mapping
- Tunneling
- Translation mechanisms



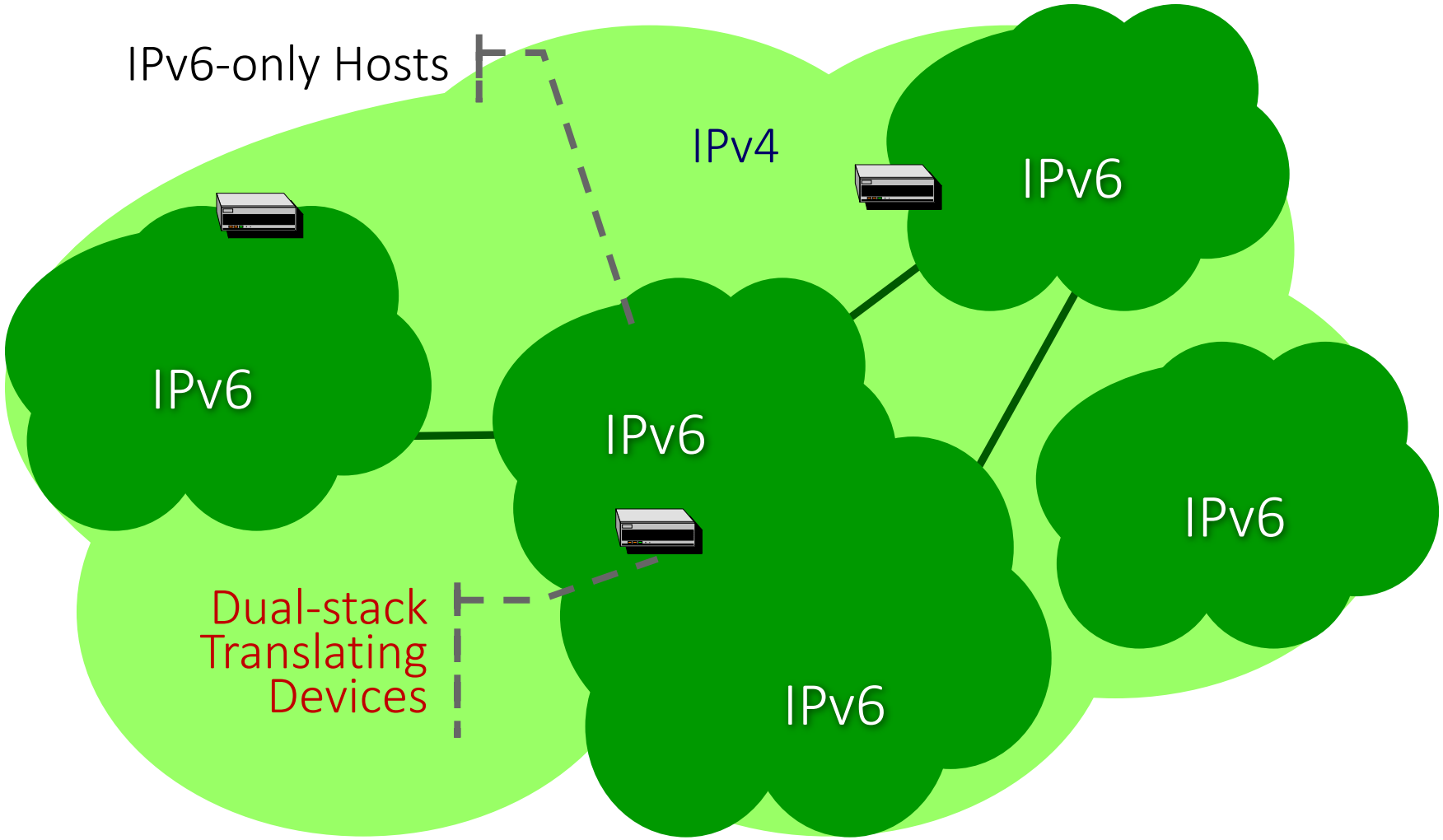
Step 1: isolated IPv6 networks



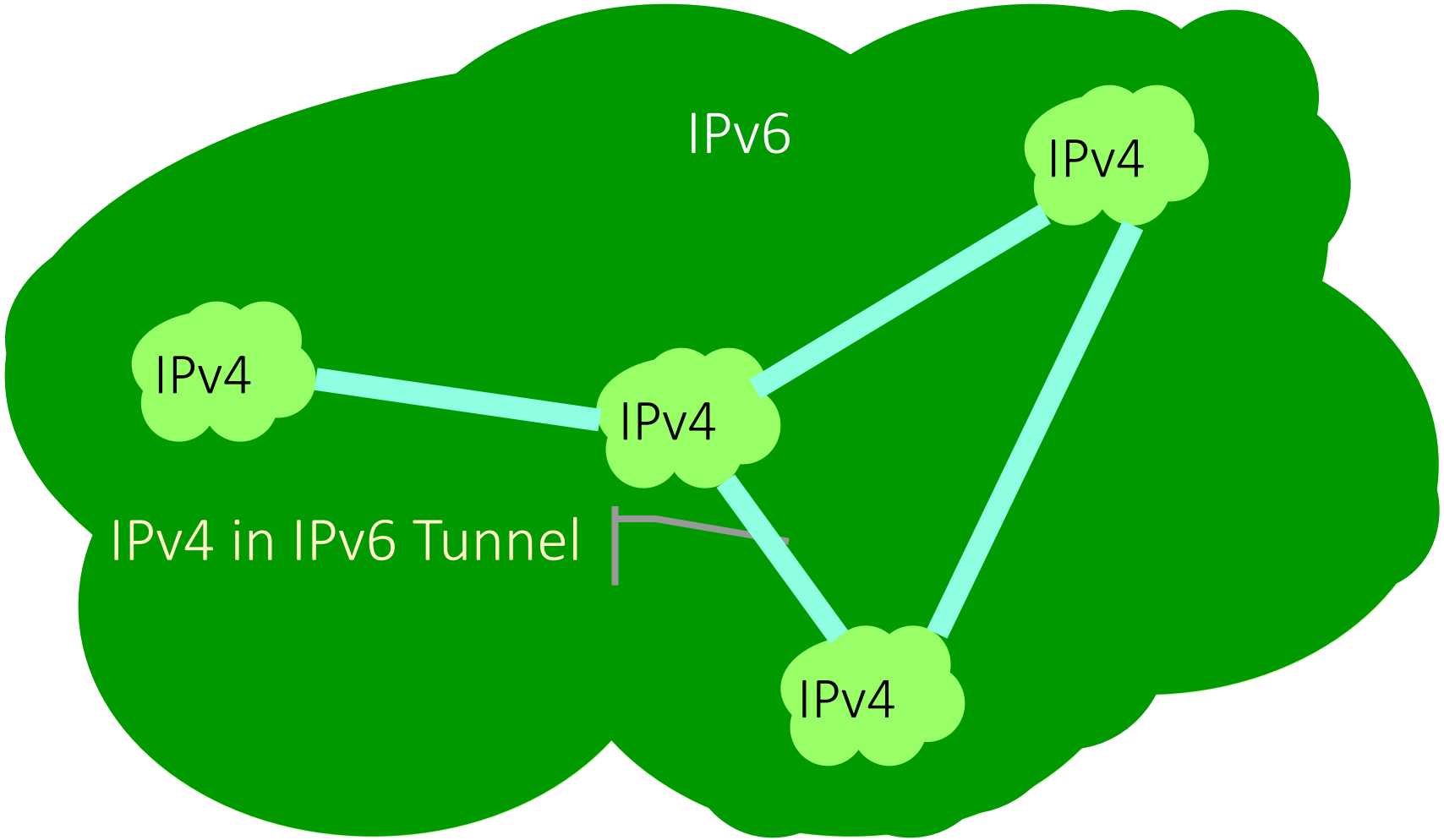
Step 2: IPv6 islands grow



Step 3: Native IPv6 connectivity



Step 4: IPv6 takes over

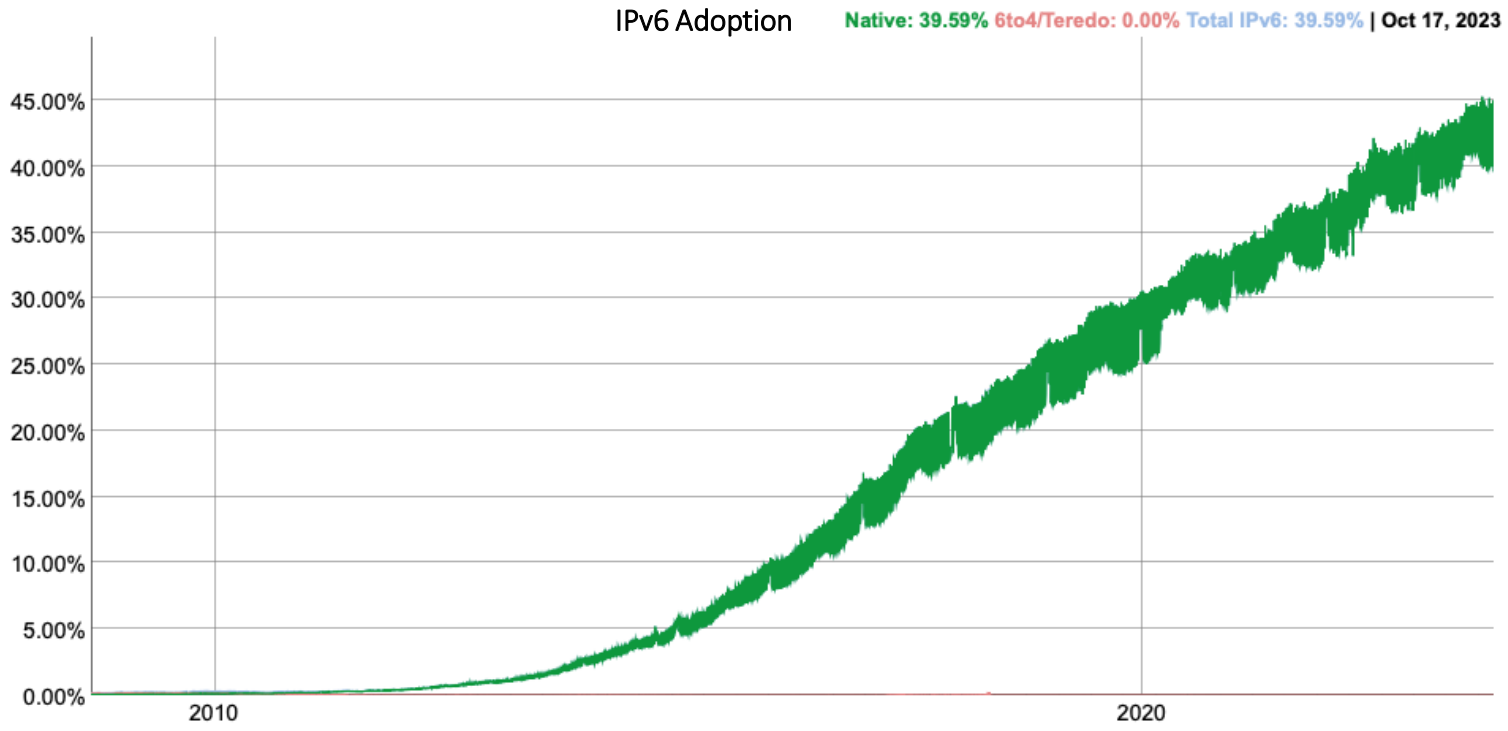


Are we ready?

- Quite so!
  - First spec RFC 1883 (1995); Internet Standard RFC 8200 (2017)
- IPv6 implemented on **routers**
  - As mature as IPv4 today, forwarded in hardware at line rate
  - Gaps remain mostly in old or low-end devices
  - Also hardware implementations (Layer-3 switches)
- IPv6 implemented in **end systems**
  - Windows (preview in 2000/XP; on by default and preferred since Vista, 2007)
  - Linux: Ubuntu, Debian, FreeBSD…
  - macOS (since Mac OS X 10.2, 2002)
  - Android and iOS: Apple requires App Store apps to work on IPv6-only networks (since 2016)

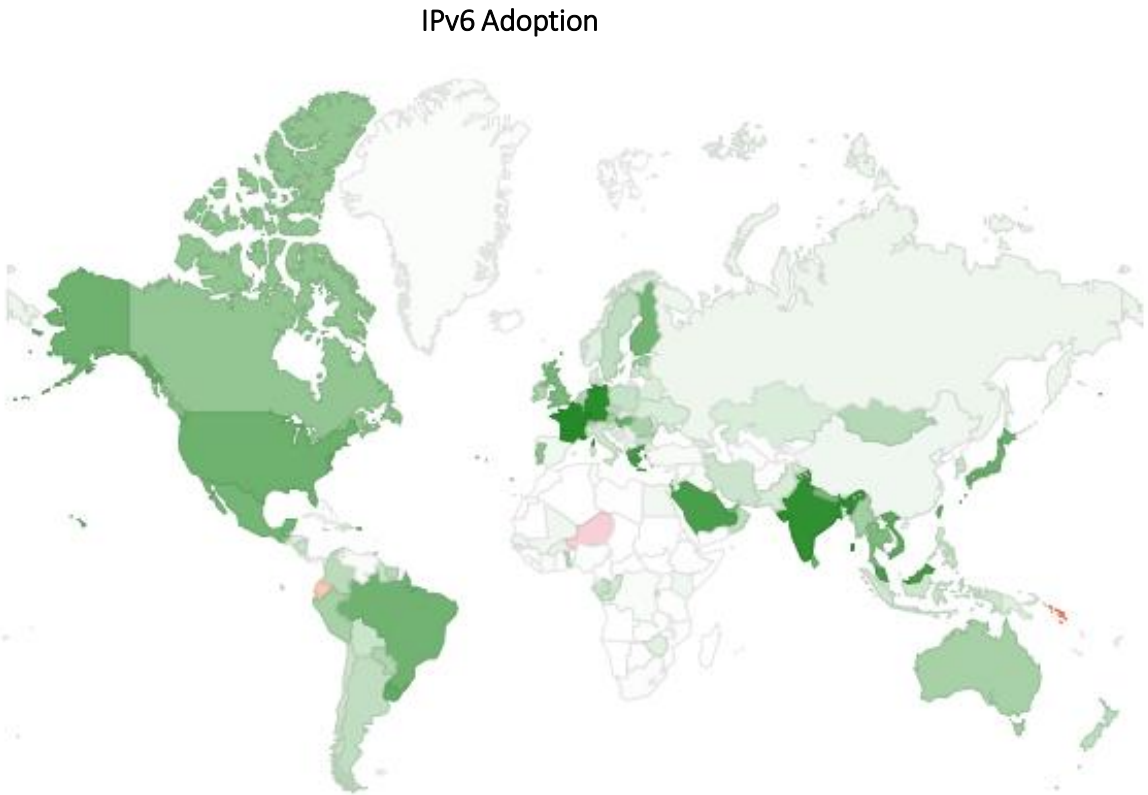
When will it happen?

- Large IPv4 install base
- As of October 2023, according to Google, the IPv6 adoption rate globally is around 45%,



When will it happen?

- Large IPv4 install base
- In 2026, according to Google, the IPv6 adoption rate globally is around 50%
  - 54% in the U.S. alone!

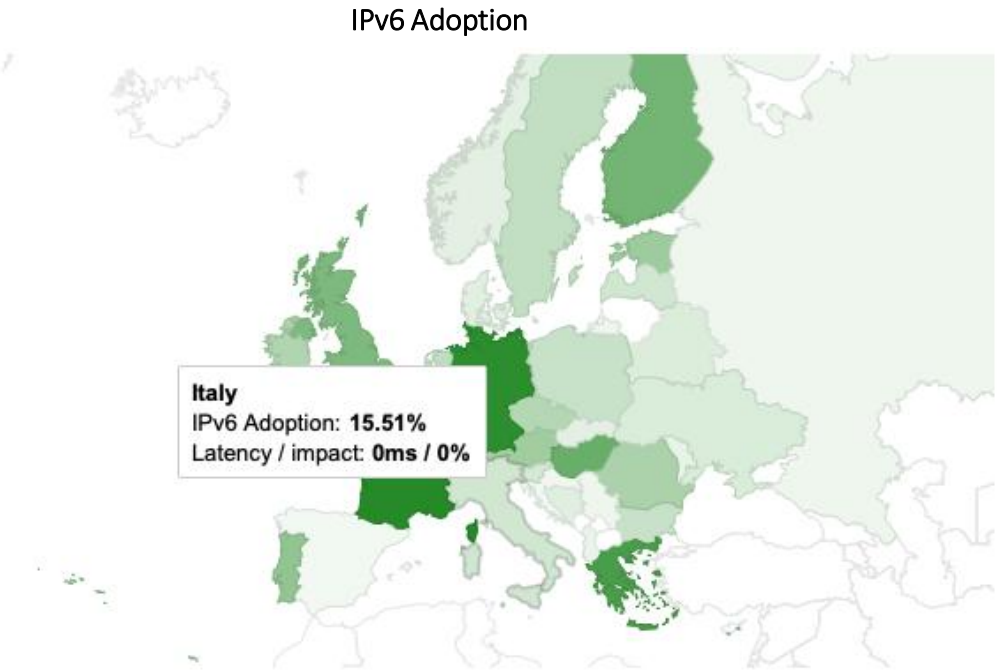


When will it happen?

- Large IPv4 install base
- In 2026, according to Google, the IPv6 adoption rate globally is around 50%
  - 54% in the U.S. alone
  - 17% in Italy, **≈86% in France (Feb. 2026), 70%+ in Germany and India**

- Only one true motivation:
  Address space depletion

- no official switch-off date
  - Will happen gradually
  - But targets exist: U.S. federal IT 80% IPv6-only by end of FY2025 (OMB M-21-07); Czech government services drop IPv4 on 6 June 2032



Assumption
(and problem to solve)

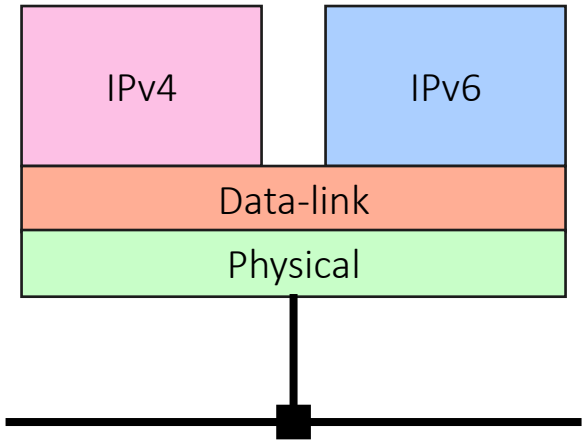
IPv4 and IPv6 will coexist

(at least for a while)



Dual Stack Approach

- Both IPv4 and IPv6 capabilities
  - All hosts and routers supporting IPv6
  - IPv4 support can be (gradually) removed (and included in new hosts) once all hosts have IPv6
- Hosts communicate natively with both
- Complete duplication of all protocol stack components
  - Routing protocols
  - Routing table
  - Access lists



Dual Stack Limitations

- It does not reduce the need for IPv4 addresses
  - Each host still needs an IPv4 address to use IPv4
- Applications have the responsibility whether to use IPv4 or IPv6
- Everything is operated twice: addressing plans, routing, firewall rules, monitoring
- In practice the OS decides

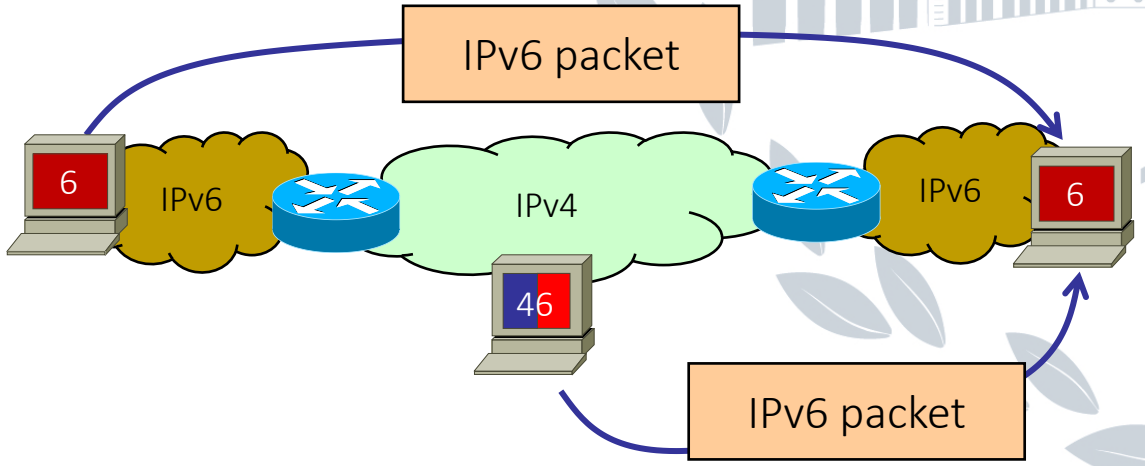
Not all hosts will be dual stack

- IPv6 hosts shall communicate with IPv6 hosts through an IPv4 network
  - Same for IPv4 hosts through an IPv6 network
- IPv6 hosts shall communicate with IPv4 hosts
  - Translation mechanisms must be used
  - Not targeting IPv4 hosts contacting IPv6 ones
    - Difficult to map the large IPv6 address space on smaller IPv4 address space

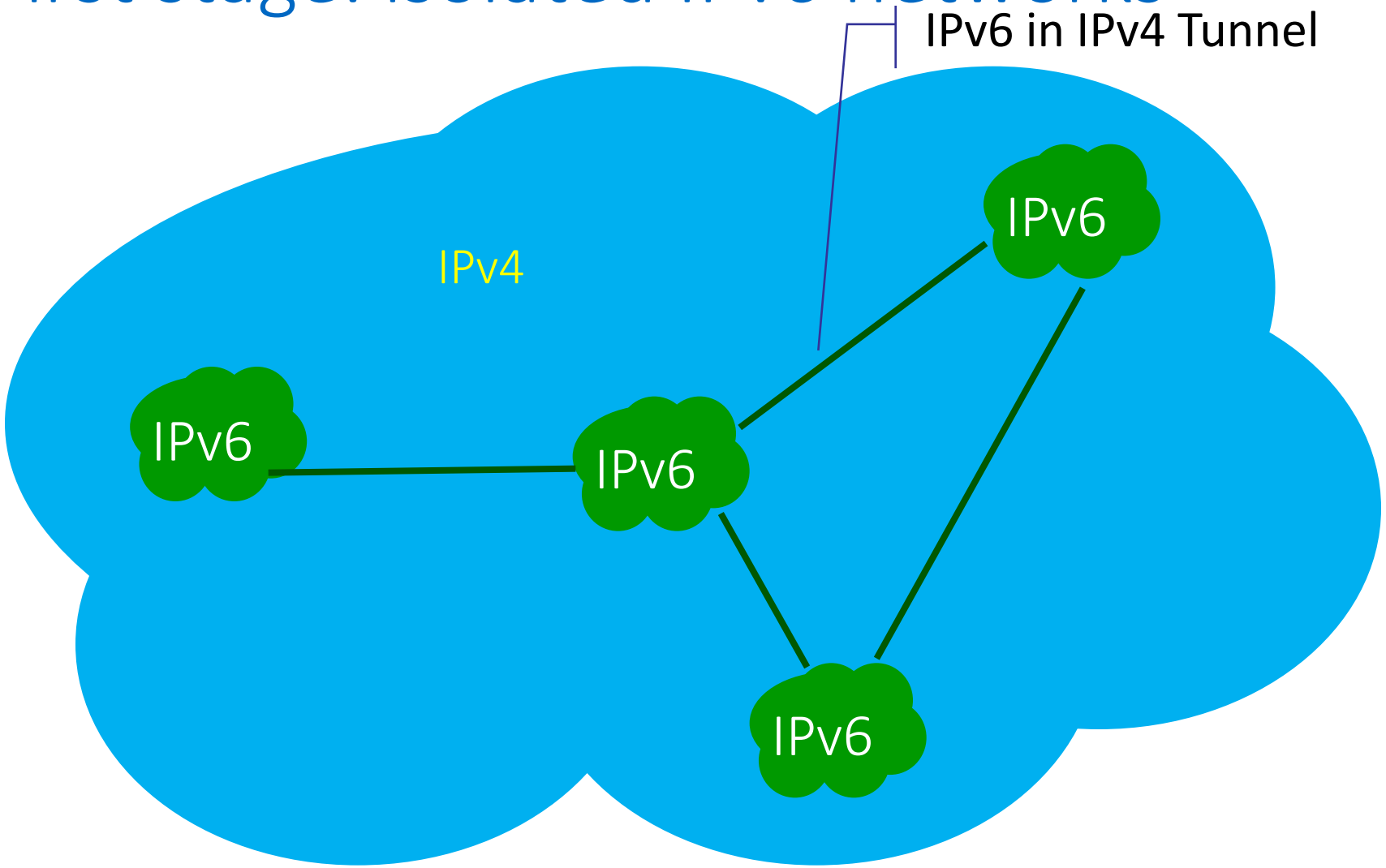


Tunneling Solutions

Dual stack and *native IPv6* hosts exchange IPv6 packets through an IPv4 network

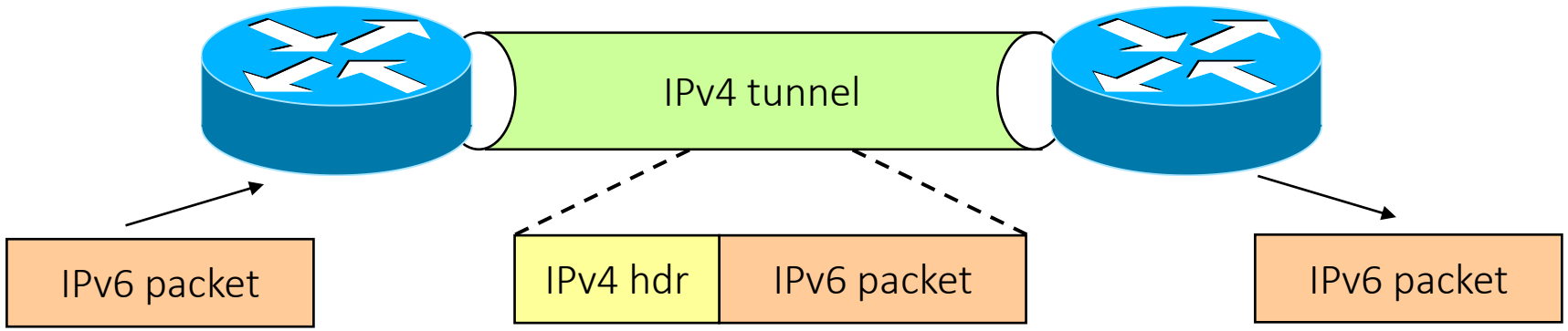


First stage: isolated IPv6 networks



Traversing a IPv4-only Network

- Tunnelling
  - Encapsulation of IP (v6) packets into IP (v4) packets
- Emulates a “direct” link among IPv6 devices



Tunnelling

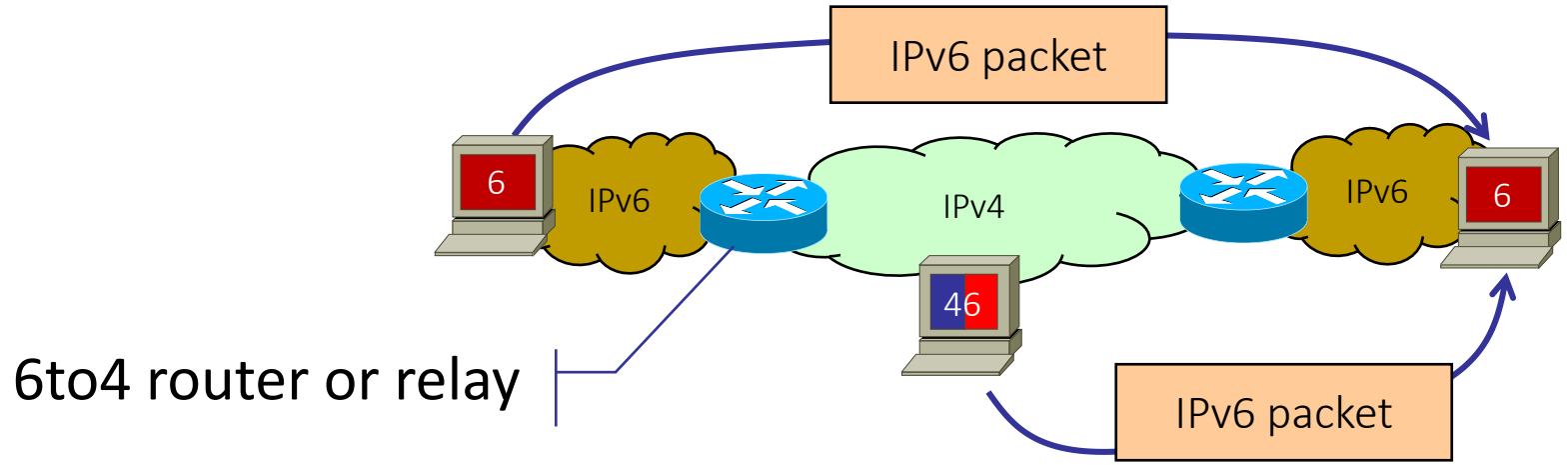
- End points: hosts and routers
- Protocols
  - GRE (Generic Routing Encapsulation)
  - IPv6 in IPv4
    - Protocol type = 41
- Set up: manual and automatic
  - IPv4-compatible addresses, 6over4 (RFC 2529), 6to4, Tunnel Broker (RFC 3053), ISATAP, Teredo

Host-centered solutions | *Network-centered solutions*

  - These solutions are either deprecated or unused today



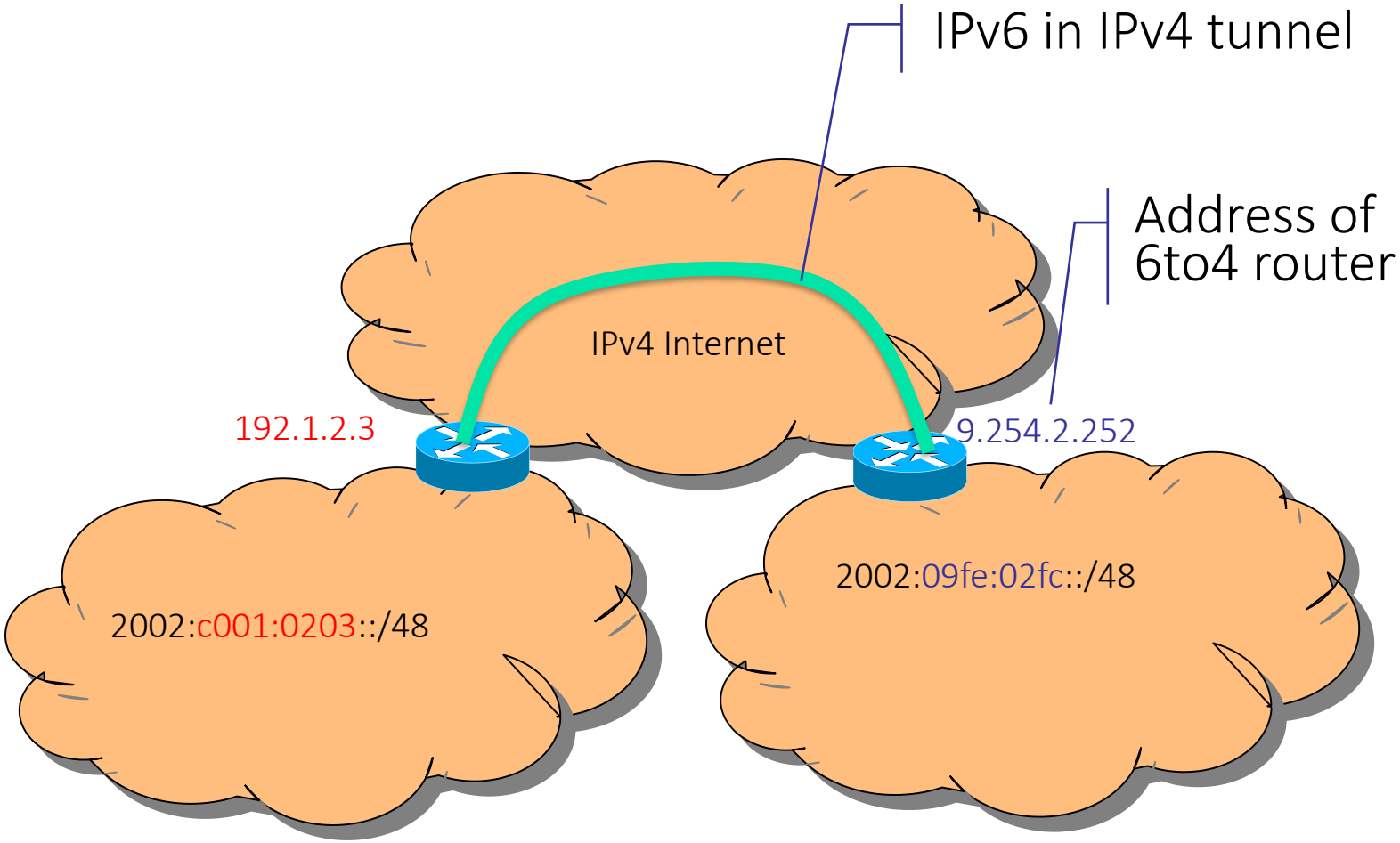
6to4

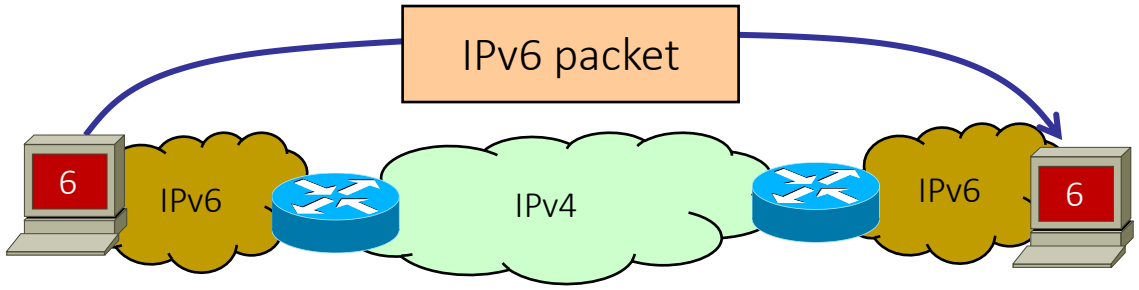


- Each 6to4 router is assigned a unique IPv6 address that begins with the prefix "*2002::/16*" followed by its 32-bit IPv4 address
- Packets sent by an IPv6 host are encapsulated in IPv4 by a 6to4 router and routed through IPv4 Internet to another 6to4 router
- The 6to4 router decapsulates it to reveal the original IPv6 packet and routes it to the destination

Basic 6to4 Scenario

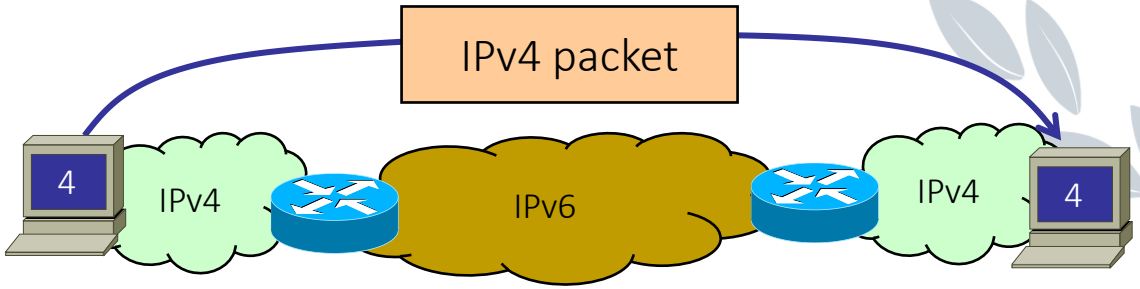
Not meant for communication between an IPv4 host and an IPv6 host





Scalable, Carrier-grade Solutions

Native IPv6 (IPv4) hosts exchange IPv6 (IPv4) packets through an IPv4 (IPv6) network

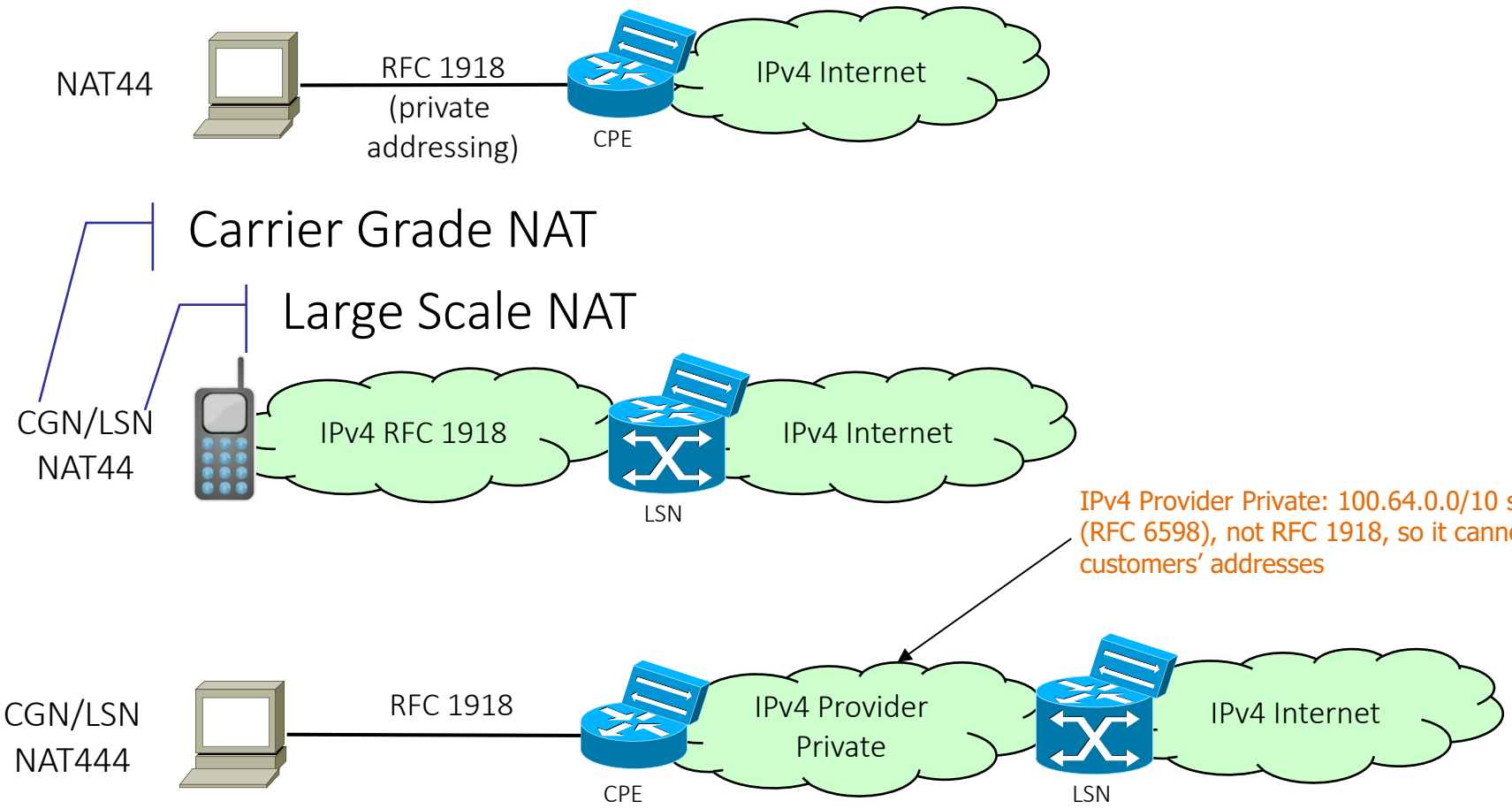


Goals

- Still need to support
  - IPv4 servers possibly communicating with IPv6 hosts
  - IPv4 clients
- Several Options
  - DS-Lite
  - A+P (Address plus Port)
    - MAP-T and MAP-E
  - NAT64
  - 464XLAT



NAT is widely used today



CPE: Customer Premises Equipment

How do we like NAT?

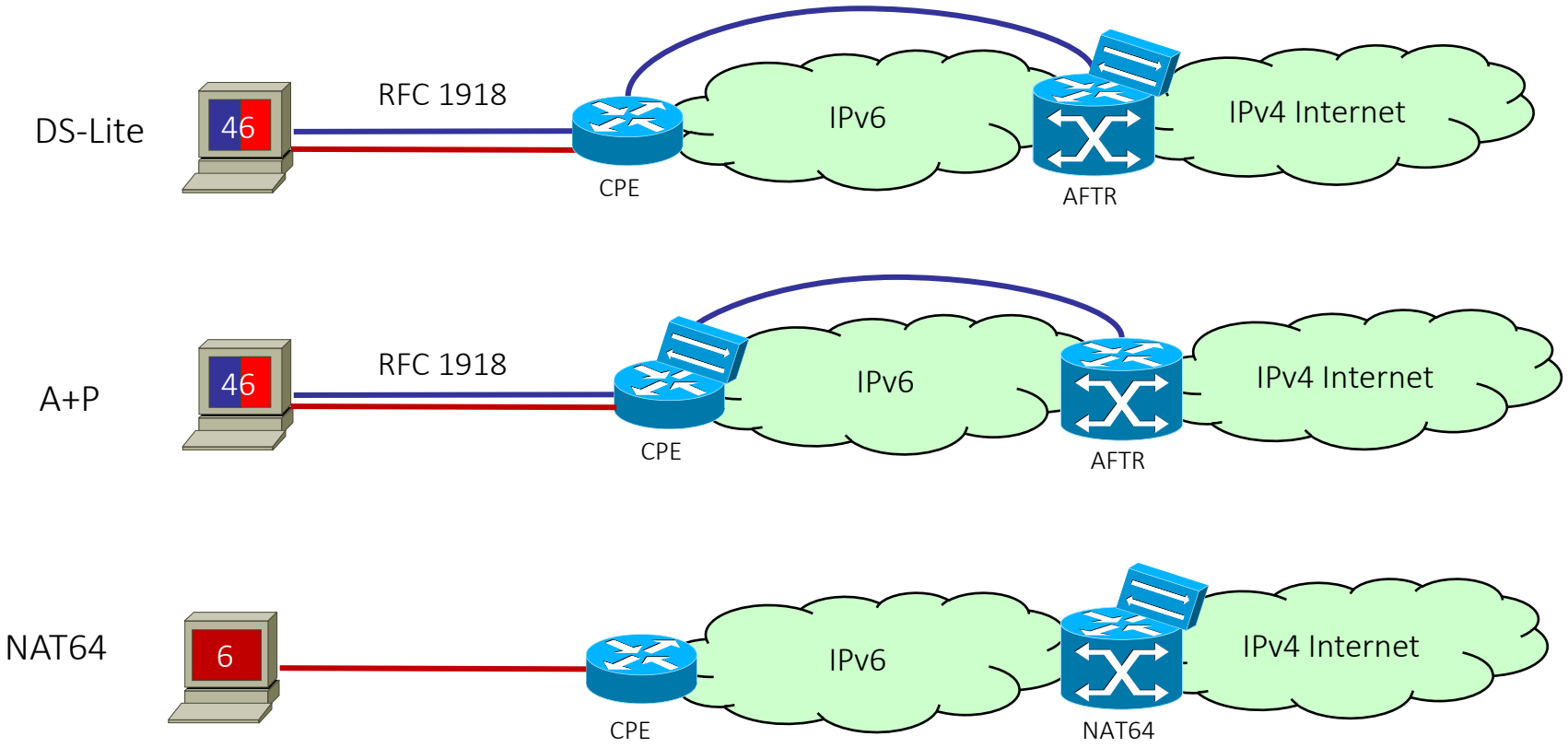
- Problematic with inbound sessions
  - E.g., servers
  - NAT + STUN/TURN may be ok for peer-to-peer sessions
- Bottleneck and single point of failure

Nevertheless

- Several (independent) cascaded instances of NAT are now very common
  - Starting from virtual machines
- Difficult to do without due to scarce addresses



Same Architecture with IPv6

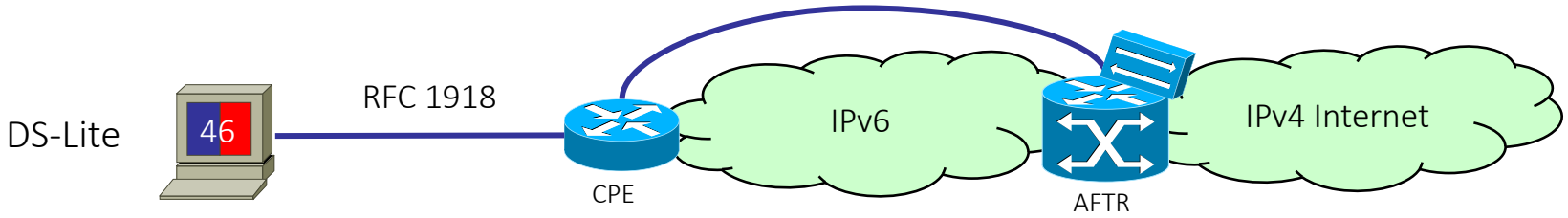


CPE: Customer Premises Equipment
AFTR: Address Family Transition Router

Note where the NAT function is located!

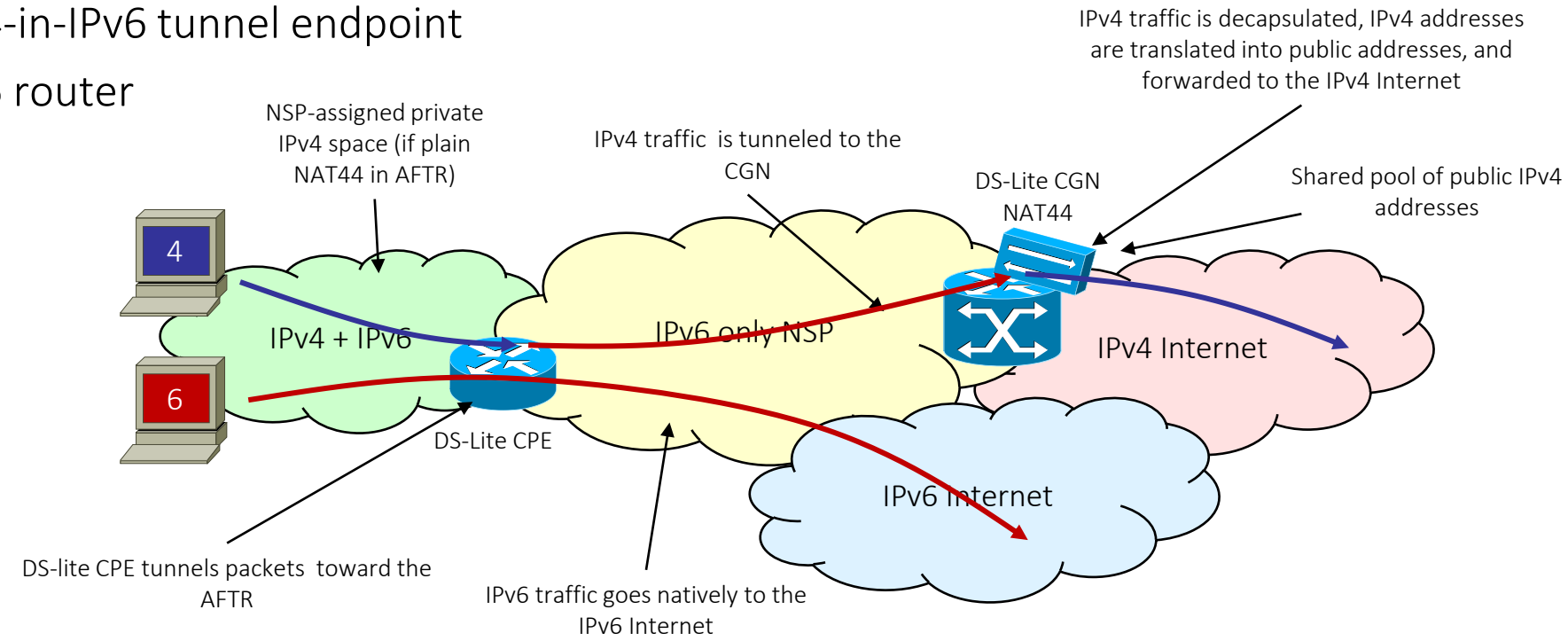
AFTR: Address Family Transition Router

- Allows IPv4 host to communicate with IPv4 hosts over an IPv6 carrier infrastructure
  - Residential host and current providers
- Features an IPv6 tunnel concentrator and possibly a large-scale NAT
- In use in DS-Lite and A+P



DS-Lite (Dual-Stack Lite)

- Dual-stack at the edge (RFC 6333)
- IPv6-only Service Provider backbone
- DS-LITE CPE: called B4 (Basic Bridging BroadBand element)
  - IPv4-in-IPv6 tunnel endpoint
  - IPv6 router



Properties

- Reduces requirement for IPv4 addresses compared to dual-stack approach
  - Dual-stack requires a public IPv4 address per customer (CPE with NAT44), or a CGN
- Extended NAT enables customer assigned (i.e., overlapping) addressing
  - IPv6 address of CPE in NAT table

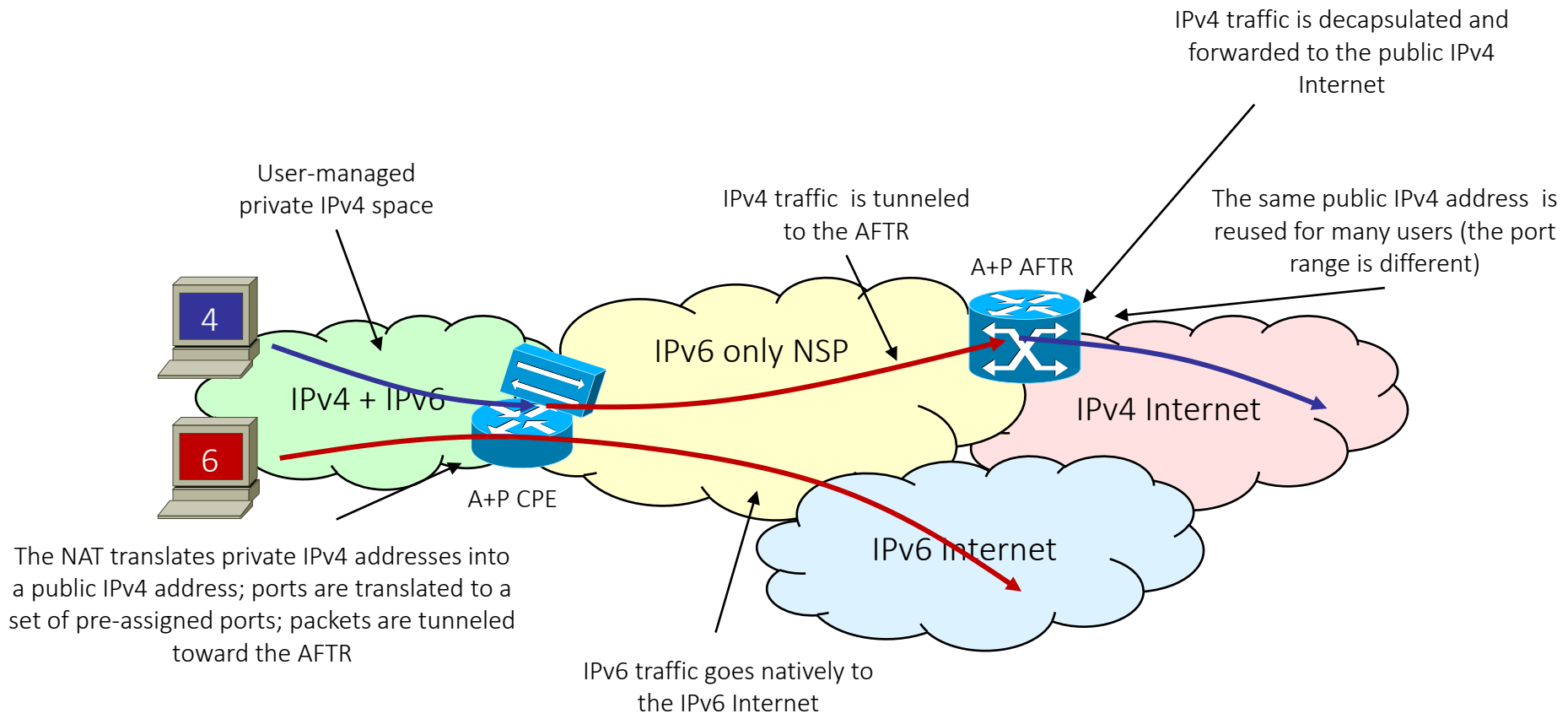
Limitations

- NAT is not under control of customer
  - Same problem as with CGN
- Problematic with servers
  - Static mapping and port forwarding cannot be configured
- AFTR keeps per-flow state, and must log address, port and time of every session for legal reasons



A+P (Address plus Port)

- NAT is under control of customer
- Ranges of TCP/UDP ports are assigned to each customer
  - Only ports used by NAT on outside



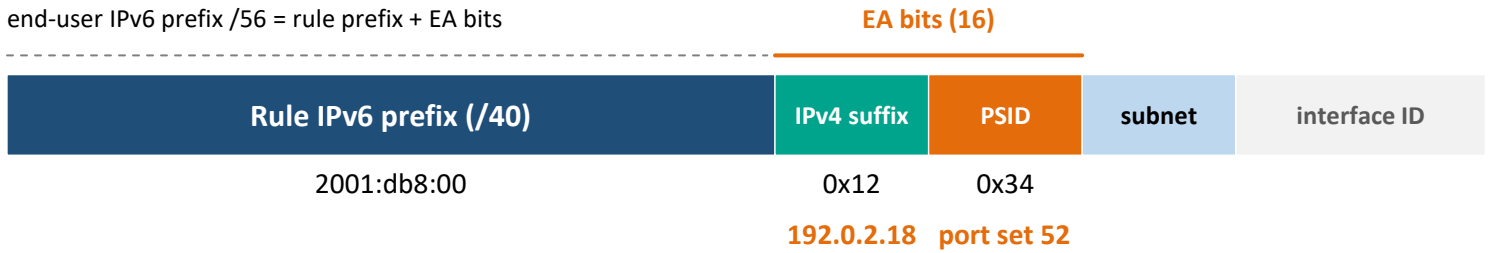
Features

- No problem with overlapping private address spaces at customers'
- Ports can be assigned automatically to CPE using the Port Control Protocol (PCP) (RFC 6887), or via DHCPv6 (RFC 7598)
  - CPE can negotiate more ports any time
- AFTR is just an IPv4-in-IPv6 (IPv6 Next Header = 4; proto-41 is IPv6-in-IPv4) tunnel terminator
  - NAT44 is no longer needed in the AFTR: one binding per customer (IPv6 address ↔ IPv4 address + port set), no per-flow state

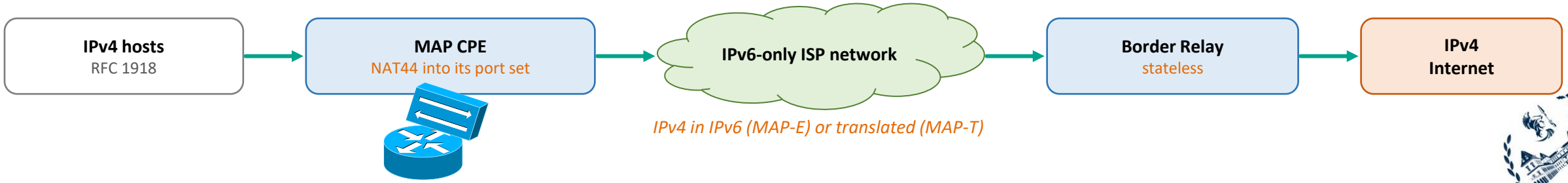


MAP-E and MAP-T: stateless A+P

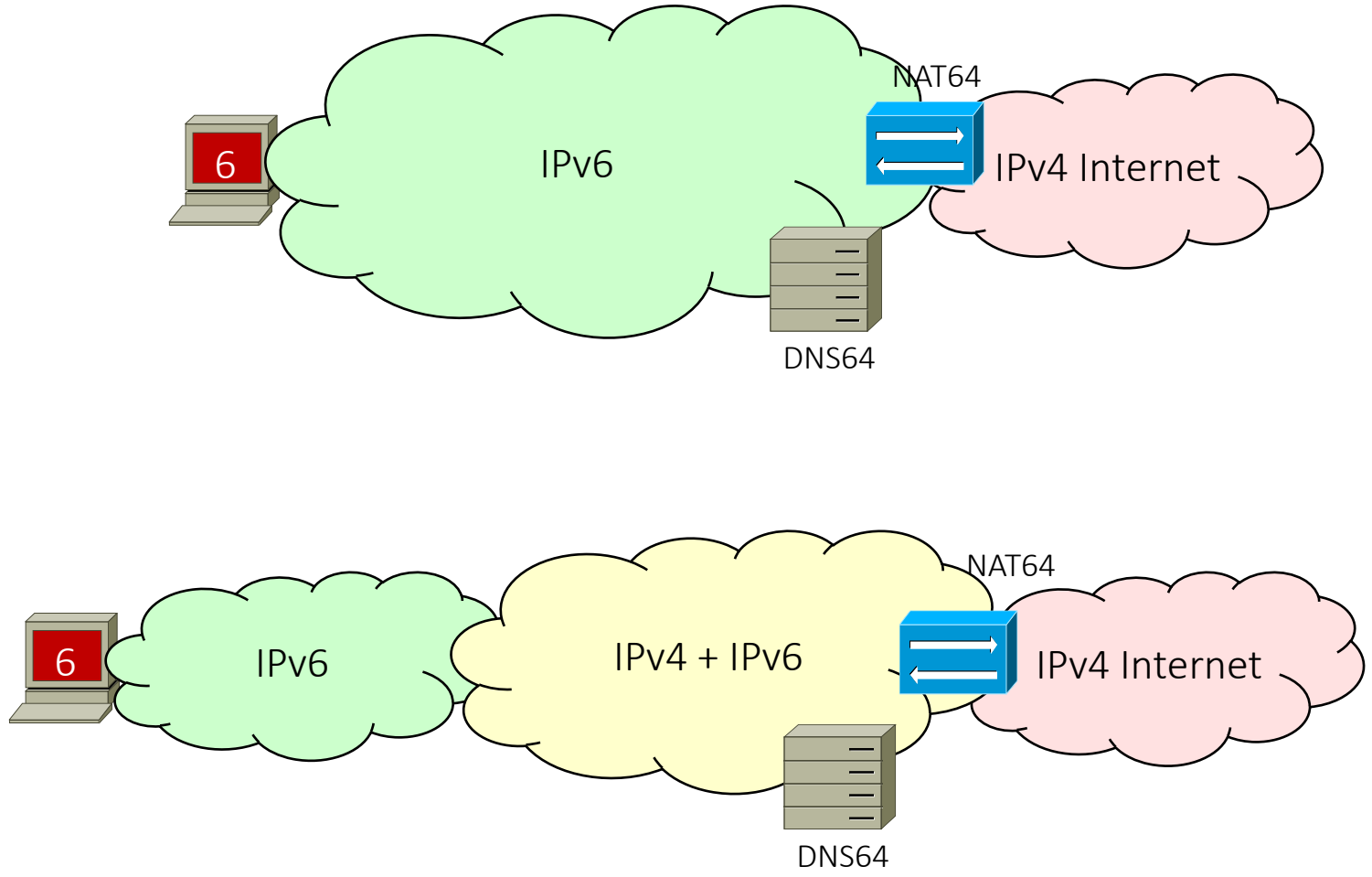
- Mapping of Address and Port: MAP-E (RFC 7597, Encapsulation), MAP-T (RFC 7599, Translation)
- Each CPE gets a shared public IPv4 address and a port set (PSID), both embedded in its delegated IPv6 prefix
  - The CPE runs NAT44 into its own port set; the Border Relay (BR) is stateless, often anycast
- MAP-E: IPv4 packet encapsulated in IPv6 (as in DS-Lite). MAP-T: IPv4 header translated into IPv6, so the IPv6 destination address carries the IPv4 destination
- No per-flow or per-customer state at the provider; MAP-E is widely used in Japan (e.g., “v6 Plus”)



Example rule: 2001:db8::/40 ↔ 192.0.2.0/24, 8 PSID bits
CPE prefix 2001:db8:12:3400::/56 → IPv4 192.0.2.18, port set 52
256 CPEs share each IPv4 address, up to 256 ports each



NAT64 + DNS64: Deployment Scenarios

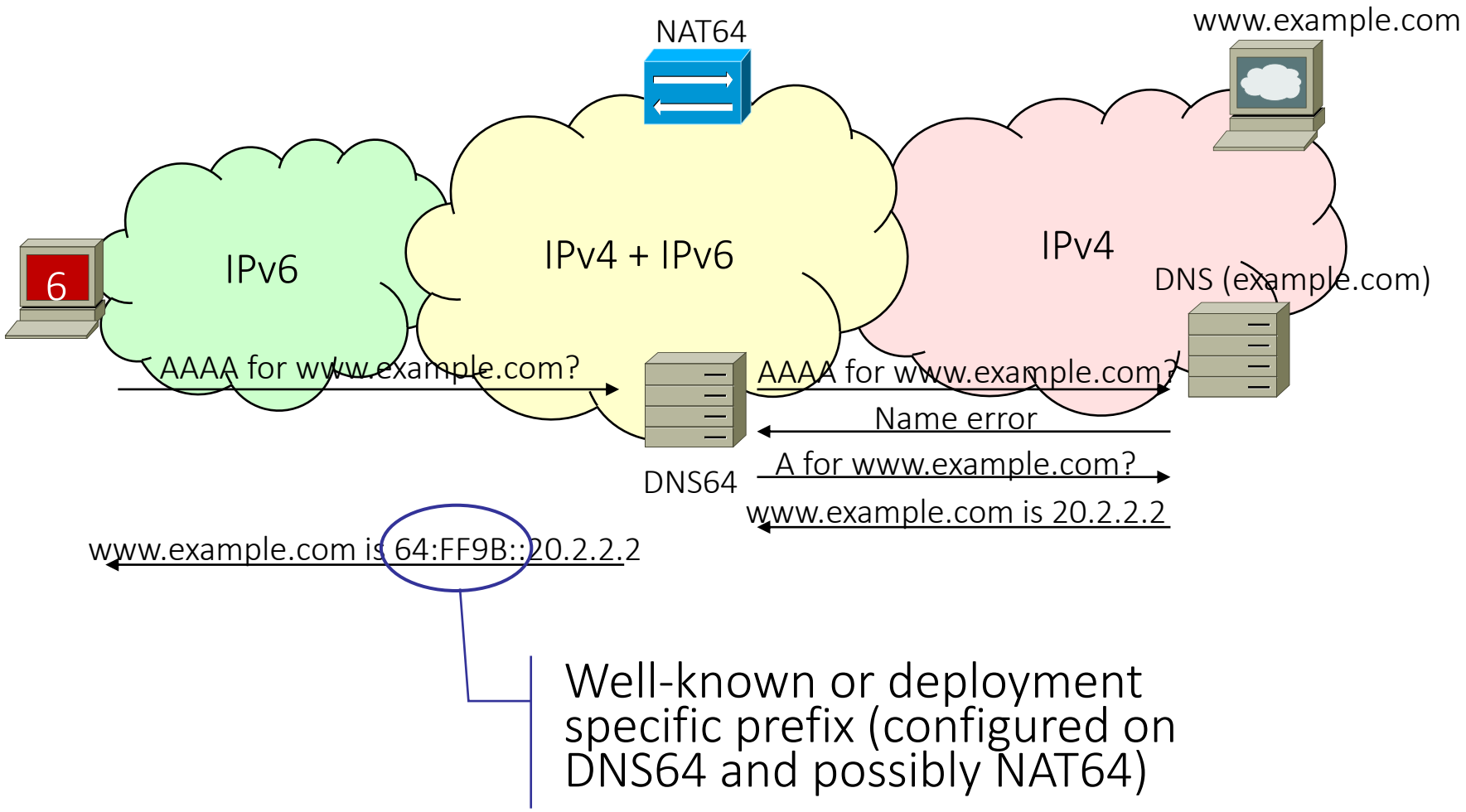


NAT64 + DNS64: Principles

- An IPv6 prefix is dedicated to mapped IPv4 addresses
  - Either well-known or network-specific
  - RFC 6052: well-known 64:ff9b::/96; local-use 64:ff9b:1::/48 (RFC 8215)
- DNS64 maps A records into AAAA using NAT64 prefix, only if the name has no native AAAA (DNS64: RFC 6147)
- NAT64 router advertises NAT64 prefix into IPv6 network to attract traffic toward IPv4 hosts
- Hosts can learn the prefix from the RA PREF64 option (RFC 8781) or by resolving ipv4only.arpa (RFC 7050)
- Stateful NAT64: RFC 6146; header translation as in SIIT (RFC 7915)

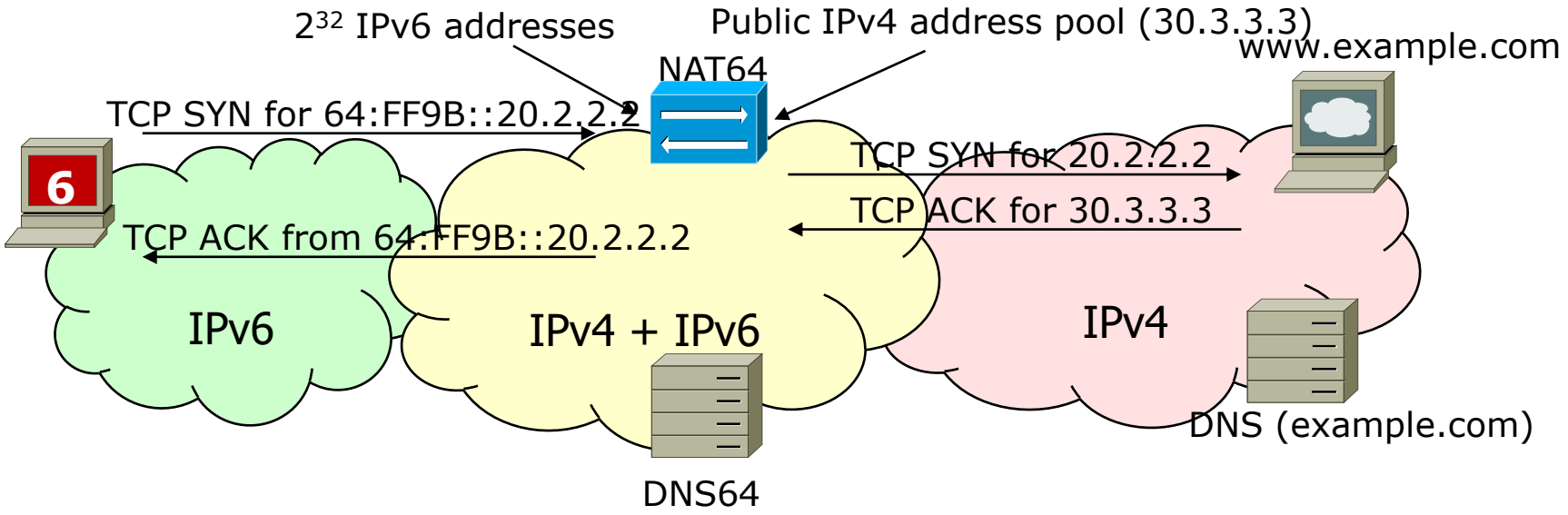


DNS64: Name Resolution



NAT64: Packet Forwarding

- NAT64 (outbound)
  - Translates IPv6 address and packet into IPv4
  - Picks a free IPv4 address/port from its pool
  - Builds NAT session entry



NAT64 + DNS64 Limitations

- Only when the DNS is involved
  - I.e., hostnames are used
  - E.g., it does not work in case the user directly specifies an IPv4 address
    - E.g., `ping 1.2.3.4`
- Same when IPv4 appear in URLs and IPv4-only applications: 464XLAT fixes this (next slide)
- DNSSEC conflict
  - In DNSSEC authoritative DNS signs record
  - But DNS64 modifies records: a validating client rejects the synthesized AAAA; it works if the DNS64 resolver validates instead (RFC 6147)



464XLAT (RFC 6877) and IPv6-only access

- Fixes the NAT64/DNS64 limits: IPv4 in URLs and IPv4-only applications keep working
- CLAT (Customer side: phone, laptop or CPE): stateless 4→6 translation (SIIT, RFC 7915) towards the NAT64 prefix
- PLAT (Provider side): an ordinary stateful NAT64
- Mobile networks run IPv6-only this way
- IPv6-mostly LANs: DHCPv4 option 108 (RFC 8925) tells capable hosts to skip IPv4; NAT64 prefix from RA PREF64 (RFC 8781). Windows 11 CLAT is rolling out to Wi-Fi/Ethernet since Sept. 2026

